# Advanced Optimization — Midterm Examination

Fall 2026 · 75 minutes · Part A: 20 true/false · Part B: 4 questions

No calculator is needed; leave fractions and square roots as they are. **Part A** (40 points): twenty statements, 2 points each. Mark each T or F. If F, give a counterexample or a one-line reason; a bare F earns 1 point, a bare T earns 2 if correct. Do Part A first.
**Part B** (60 points, 15 each): Questions 1 and 2 ask for explanations; Questions 3 and 4 ask you to work a small example and then to explain what it shows. The credit is for the explanation in your own words; a quoted passage without explanation earns none.

## Part A — True or false

### Chapter 2

**A1.** If $x^*$ is a strict local minimizer of $f$, then it is an isolated local minimizer.

**A2.** If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semidefinite, then $x^*$ is a local minimizer.

**A3.** Newton's method is invariant under a linear change of variables $x = Sz$ ($S$ nonsingular): the iterates in $z$ map to the iterates in $x$. Steepest descent is not.

**A4.** The sequence $x_k = 1 + k^{-k}$ converges to 1 Q-superlinearly but not Q-quadratically.

### Chapter 3

**A5.** Let $f$ be smooth and bounded below along the ray $\{x_k + \alpha p_k : \alpha > 0\}$, with $p_k$ a descent direction. Then for any $0 < c_1 < c_2 < 1$ there exist step lengths satisfying the strong Wolfe conditions.

**A6.** Zoutendijk's theorem implies that steepest descent with a Wolfe line search converges to a local minimizer.

**A7.** Let $x_k \to x^*$ with $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$, and let $p_k$ be the Newton direction. With a Wolfe line search with $c_1 \le \frac{1}{2}$, the unit step $\alpha_k = 1$ satisfies the Wolfe conditions for all sufficiently large $k$.

**A8.** For $f(x) = \frac{1}{2} x^{\mathsf{T}} Q x$ with $Q \succ 0$, steepest descent with exact line searches reaches the minimizer in at most $n$ iterations.

### Chapter 4

**A9.** In Algorithm 4.1, the trust-region radius is increased whenever $\rho_k > \frac{3}{4}$.

**A10.** The Cauchy point always lies on the boundary of the trust region.

**A11.** If $B_k$ is indefinite, the exact solution of the trust-region subproblem lies on the boundary $\|p\| = \Delta_k$.

**A12.** The dogleg method for the trust-region subproblem requires $B_k$ to be positive definite.

### Chapter 5

**A13.** Applied to $Ax = b$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, the conjugate gradient method needs exactly $n$ iterations to reach the solution in exact arithmetic.

**A14.** In the conjugate gradient method the residuals $r_0, r_1, \dots$ are mutually orthogonal and the directions $p_0, p_1, \dots$ are mutually conjugate.

**A15.** The $k$-th CG iterate $x_k$ minimizes $\phi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ over the whole affine set $x_0 + \operatorname{span}\{p_0, \dots, p_{k-1}\}$, not only along the last direction $p_{k-1}$.

**A16.** The number of CG iterations needed to reach a given accuracy is determined by the condition number $\kappa(A)$.

**Chapter 6**

**A17.** If $f$ is strongly convex, then $s_k^\mathsf{T} y_k > 0$ for any two points $x_k, x_{k+1}$, whatever line search is used.

**A18.** If $B_k$ is positive definite and $s_k^\mathsf{T} y_k > 0$, then the BFGS update $B_{k+1}$ is positive definite.

**A19.** If $B_k$ is positive definite and $s_k^\mathsf{T} y_k > 0$, then the SR1 update $B_{k+1}$ is positive definite.

**A20.** BFGS with a Wolfe line search satisfies $\liminf \|\nabla f_k\| = 0$ for every twice continuously differentiable $f$ that is bounded below.

# Part B

## Question 1 (Chapter 3)

What is the Armijo (sufficient decrease) condition? What are the Wolfe conditions? Why is the Armijo condition with backtracking not always enough — when is it enough, and when is it not?

## Question 2 (Chapters 3–6)

In unconstrained minimization the objective is modeled locally by a quadratic $m_k(p) = f_k + \nabla f_k^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B_k p$. Positive definiteness of $B_k$ (or of the Hessian) is often assumed, but not always needed.

(A) List the methods of Chapters 3–6 that require $B_k$ to be positive definite, and say for each what breaks if it is not.

(B) List the methods that do not require it, and explain in each case how that is possible.

## Question 3 (Chapter 4)

Consider the model $m(p) = g^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B p$ with $g = (1, 0)^\mathsf{T}$ and $B = \mathrm{diag}(1, -1)$, and the trust region $\|p\| \le 1$. A line search method would first make $B$ positive definite (e.g. $B + \tau I$ with $\tau > 1$) and then search along $-(B + \tau I)^{-1} g$: find the best model value such a method can reach, for any $\tau$ and any step length within the trust region. Then solve the trust-region subproblem exactly using Theorem 4.1 (note what happens to the condition $\|p(\lambda)\| = \Delta$), give its model value, and explain what the trust-region step does with the negative curvature that the line search step cannot do.

## Question 4 (Chapter 5)

Let $A = \mathrm{diag}(1, 3)$, $b = 0$, $x_0 = (3, 1)^\mathsf{T}$, with residual $r_k = A x_k - b$; the solution is $x^* = 0$. Starting from $p_0 = -r_0$, take the exact line search step to $x_1$ and compute $r_1$. Then find the direction of the form $p_1 = -r_1 + \beta p_0$ that is conjugate to $p_0$ (i.e. $p_0^\mathsf{T} A p_1 = 0$), take the exact line search step along it, and verify that you arrive at $x^*$. For comparison, compute where exact-line-search steepest descent ($p_1 = -r_1$) lands instead, and explain why it misses $x^*$ while the conjugate direction does not.